# Divide-and-Conquer Technique:
## Finding Maximum & Minimum

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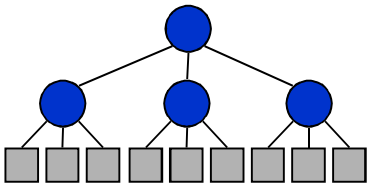
## Divide-and-Conquer

- Divide-and-Conquer is a general algorithm design paradigm:
  - Divide the problem into a number of subproblems that are smaller instances of the same problem
  - Conquer the subproblems by solving them recursively
  - Combine the solutions to the subproblems into the solution for the original problem
- The base case for the recursion are subproblems of constant size
- Analysis can be done using **recurrence equations**



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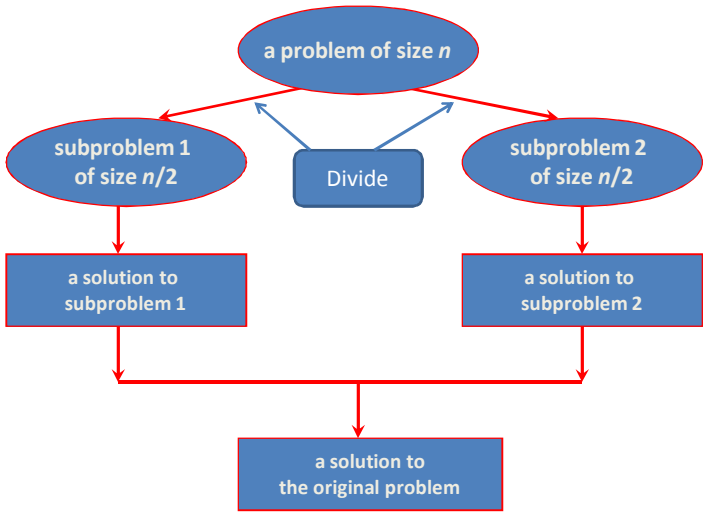
# Divide-and-Conquer

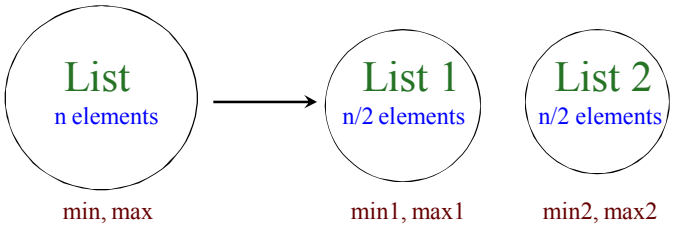





# Finding Maximum and Minimum

- *Input:* an array $A[1..n]$ of $n$ numbers
- *Output:* the maximum and minimum value

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | 13 | -13 | -25 | 20 | -3 | -16 | -23 | 18 | 20 | -7 | 12 | -5 | -22 | 15 | -4 | 7 |




**min = MIN ( min1, min2 )**
**max = MAX ( max1, max2 )**

## Finding Maximum and Minimum

*The straightforward algorithm:*

```
max ← min ← A [1];
for i ← 2 to n do
    if (A [i] > max) then max ← A [i];
    if (A [i] < min) then  min ← A [i];
```

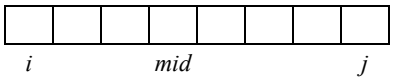
No. of comparisons: $2(n-1)$

Running Time $T(n) = O(n)$



## Finding Maximum and Minimum

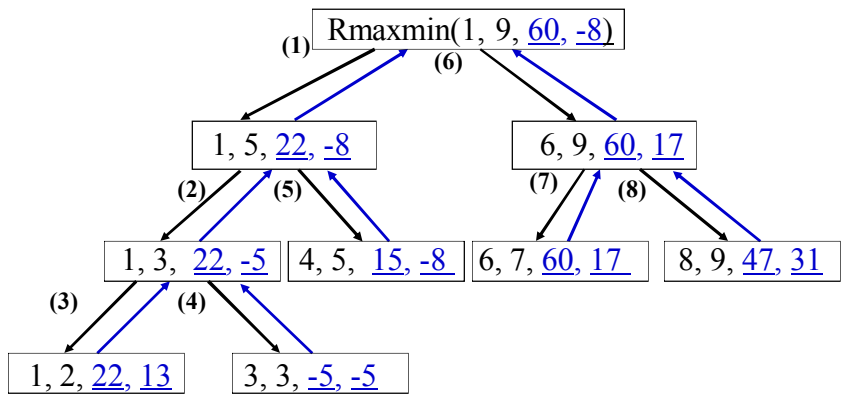
*The Divide-and-Conquer algorithm:*




```
procedure Rmaxmin (i, j, fmax, fmin);
    begin
        case:
            i = j:          fmax ← fmin ← A[i];
            i +1 = j:       if A[i] < A[j] then   { fmax ← A[j];
                                                    fmin ← A[i]; }
                            else { fmax ← A[i];
                                   fmin ←  A[j]; }
            else:           mid ← (i + j)/2;
                            call Rmaxmin (i, mid, gmax, gmin);
                            call Rmaxmin (mid+1, j, hmax, hmin);
                            fmax ← MAX (gmax, hmax);
                            fmin ← MIN (gmin, hmin);
        end
    end;
```

## Finding Maximum and Minimum

| Index: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| Array: | 22 | 13 | -5 | -8 | 15 | 60 | 17 | 31 | 47 |






## Finding Maximum and Minimum

The recurrence for the worst-case running time $T(n)$ is

$$T(n) = \begin{cases} O(1) & \text{if } n = 1 \text{ or } 2 \\ 2T(n/2) + O(1) & \text{if } n > 2 \end{cases}$$

**equivalently**

$$T(n) = \begin{cases} b & \text{if } n = 1 \text{ or } 2 \\ 2T(n/2) + b & \text{if } n > 2 \end{cases}$$

By solving the recurrence, we get

$$T(n) = O(n)$$